

MATH 350-2 Advanced Calculus

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Outline

- 1 Real Analysis Lecture 19
 - Differentiation
 - Mean Value Theorem

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Differentiability

Definition

A real-valued function $f(x)$ defined on an open interval (a, b) is **differentiable** at $c \in (a, b)$ if

$$\lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c} \text{ exists.}$$

In this case the limit is called the derivative of f at c , and denote by $f'(c)$.

Chord function

Theorem (Helper Theorem)

Let f be a real-valued function on (a, b) . Then f is differentiable at $c \in (a, b)$ if and only if there exists a function f^ on (a, b) which is continuous at c and satisfies*

$$f(x) - f(c) = (x - c)f^*(x).$$

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The function $f^*(x)$ is the **chord slope function**.

Chord function

Proof.

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Suppose $f(x)$ is differentiable at c .

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Suppose $f(x)$ is differentiable at c . Define

$$f^*(x) = \begin{cases} \frac{f(x)-f(c)}{x-c}, & x \neq c \\ f'(c), & x = c \end{cases}$$

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Proof.

Suppose $f(x)$ is differentiable at c . Define

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Then $\lim_{x \rightarrow c} f^*(x) = f'(c)$, so f^* is continuous at c .

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Conversely, suppose f^* exists and is continuous at c .

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$$\lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c} = \lim_{x \rightarrow c} f^*(x) = f^*(c).$$

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Conversely, suppose f^* exists and is continuous at c .
Then

$$\lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c} = \lim_{x \rightarrow c} f^*(x) = f^*(c).$$

In particular f is differentiable at c and $f'(x) = f^*(c)$. □

Differentiable implies continuous

Theorem

*Suppose $f(x)$ is a real valued function on (a, b) and is differentiable at $c \in (a, b)$.
Then f is continuous at c .*

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By the Helper Theorem, there exists a function f^* on (a, b) which is continuous at c with

$$f(x) = (x - c)f^*(x).$$

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Proof.

By the Helper Theorem, there exists a function f^* on (a, b) which is continuous at c with

$$f(x) = (x - c)f^*(x).$$

Since $(x - c)$ and f^* are both continuous at c , so is $f(x)$. □

Additivity of derivatives

Theorem

Let f, g be real functions on (a, b) which are differentiable at $c \in (a, b)$.

$$(f \pm g)'(c) = f'(c) \pm g'(c)$$

Additivity of derivatives

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By the Helper Theorem

$$f(x) - f(c) = (x - c)f^*(x), \quad g(x) - g(c) = (x - c)g^*(x)$$

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$$f(x) - f(c) = (x - c)f^*(x), \quad g(x) - g(c) = (x - c)g^*(x)$$

Therefore

$$(f \pm g)(x) - (f \pm g)(c) = (x - c)(f^*(x) \pm g^*(x)).$$

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Since f^* and g^* are continuous at c , so is $f^* \pm g^*$.

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Since f^* and g^* are continuous at c , so is $f^* \pm g^*$.

The Helper Theorem implies that $g \pm f$ is differentiable and

$$(f \pm g)'(c) = f'(c) \pm g'(c)$$



Product rule

Theorem

Let f, g be real functions on (a, b) which are differentiable at $c \in (a, b)$.

$$(fg)'(c) = f'(c)g(c) + f(c)g'(c)$$

Product rule

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Therefore

$$\begin{aligned}(fg)(x) - (fg)(c) &= f(x)g(x) - f(c)g(c) \\&= f(x)g(x) - f(x)g(c) + f(x)g(c) - f(c)g(c) \\&= f(x)(g(x) - g(c)) + (f(x) - f(c))g(c) \\&= (x - c)[f(x)g^*(x) + f^*(x)g(c)]\end{aligned}$$

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Helper Theorem says:

$$(fg)'(c) = f(c)g^*(c) + f^*(c)g(c) = f(c)g'(c) + f'(c)g(c).$$

Quotient rule

Theorem

Let f, g be real functions on (a, b) which are differentiable at $c \in (a, b)$ with $g(c) \neq 0$.

$$(f/g)'(c) = \frac{f'(c)g(c) - f(c)g'(c)}{g(c)^2}$$

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Chain rule

Theorem

Let f be a real function on an interval $I = (a, b)$, and let g be a real function on $f(I)$, and consider the composition

$$h = g \circ f : (a, b) \rightarrow \mathbb{R}.$$

Suppose $c \in (a, b)$ and $f(c)$ is an interior point of $f(I)$. If f is differentiable at c and g is differentiable at $f(c)$, then h is differentiable at c and

$$h'(c) = (g \circ f)'(c) = g'(f(c))f'(c).$$

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Local maxima and minima

Definition

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$$f(x) \leq f(c) \quad \text{for all } x \in S.$$

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It has **local maximum at** $a \in S$ if there exists an $r > 0$ such that

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Absolute and local minima are defined similarly.

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Definition

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Theorem

Let f be a real-valued function on an open interval (a, b) and suppose that f has a local maximum or local minimum at $c \in (a, b)$. Then c is a critical point of f .

Critical points

Proof.

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By definition, there exists $r > 0$ such that

$$f(x) \leq f(c) \quad \text{for all } x \in (c - r, c + r) \cap (a, b).$$

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To obtain a contradiction, suppose that $f'(c) \neq 0$.

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To obtain a contradiction, suppose that $f'(c) \neq 0$.

Then by the Helper Theorem, there is a function $f^*(x)$ defined on (a, b) and continuous at $c \in (a, b)$ with

$$f(x) - f(c) = (x - c)f^*(x).$$



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Since f is continuous, we may choose $\delta > 0$ such that

$$|x - c| < \delta \Rightarrow |f^*(x) - f^*(c)| < |f^*(c)|/2.$$

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It follows that $f^*(x) > |f^*(c)|/2$ for all $x \in (a, b)$ with $|x - c| < \delta$.

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It follows that $f^*(x) > |f^*(c)|/2$ for all $x \in (a, b)$ with $|x - c| < \delta$.

However,

$$f(c \pm \delta/2) = f(c) \pm \frac{1}{2}\delta f^*(c \pm \delta/2),$$

one of these will be larger than $f(c)$.

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Proof.

Moreover, $f^*(c) = f'(c) \neq 0$.

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It follows that $f^*(x) > |f^*(c)|/2$ for all $x \in (a, b)$ with $|x - c| < \delta$.

However,

$$f(c \pm \delta/2) = f(c) \pm \frac{1}{2} \delta f^*(c \pm \delta/2),$$

one of these will be larger than $f(c)$. Contradiction!



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Then there exists $c \in (a, b)$ with $f'(c) = 0$.

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By the Extreme Value Theorem, $f(x)$ has an absolute maximum on $[a, b]$, as well as an absolute minimum on $[a, b]$. If both of these occur at the endpoints, then the absolute max is the absolute min, forcing f to be constant.

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By the Extreme Value Theorem, $f(x)$ has an absolute maximum on $[a, b]$, as well as an absolute minimum on $[a, b]$. If both of these occur at the endpoints, then the absolute max is the absolute min, forcing f to be constant. Then $f'(c) = 0$ for every $c \in (a, b)$.

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By the Extreme Value Theorem, $f(x)$ has an absolute maximum on $[a, b]$, as well as an absolute minimum on $[a, b]$. If both of these occur at the endpoints, then the absolute max is the absolute min, forcing f to be constant.

Then $f'(c) = 0$ for every $c \in (a, b)$.

Otherwise at least one of the points happens inside, so $f(x)$ has a local max or local min at a value c inside (a, b) .

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Otherwise at least one of the points happens inside, so $f(x)$ has a local max or local min at a value c inside (a, b) .

The point c is a critical point, and since f is differentiable at c , we must have $f'(c) = 0$. □

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Then there exists $c \in (a, b)$ with

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

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Proof.

Consider the function

$$g(x) = f(x) - \left(\frac{f(b) - f(a)}{b - a} \right) (x - a) + f(a).$$

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- differentiable at each point in (a, b) ;
- and satisfies $g(a) = g(b)$.

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Then $g(x)$ is

- continuous on $[a, b]$;
- differentiable at each point in (a, b) ;
- and satisfies $g(a) = g(b)$.

By Rolle's Theorem, there exists $c \in (a, b)$ with

$$0 = g'(c) = f'(c) - \frac{f(b) - f(a)}{b - a}.$$